

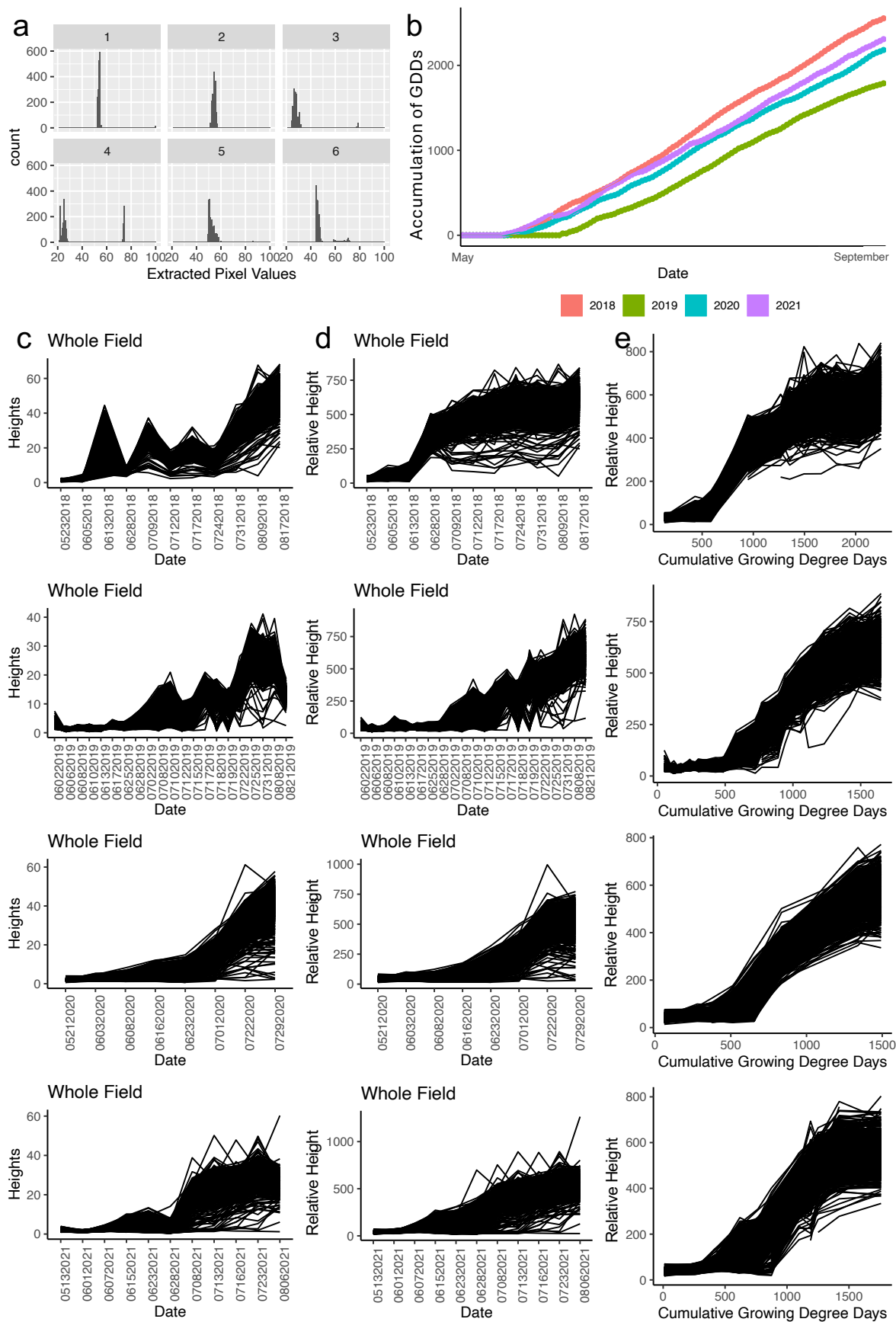


Figure S1. Normalization of extracted plant height values. (a) Example values extracted from GCP bounding boxes to identify height values for the ground and tops of the GCPs (06022019). (b) Accumulation of growing degree days (GDDs) throughout the growing season for each year. Extracted plant height values across the growing season for all plots with (c) raw data (before any cleaning), (d) normalized data (before cleaning but normalized across dates using GCP heights), and (e) clean data (normalized data with erroneous plots removed before analysis). Dates on the x-axis are in the form MMDDYYYY.

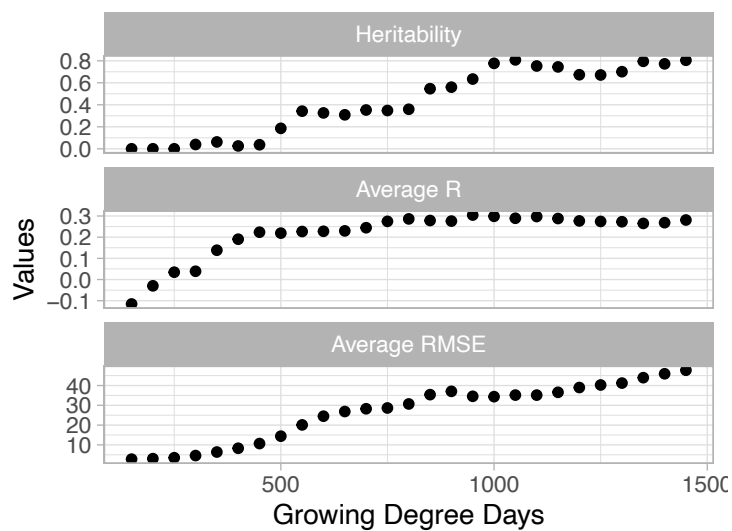


Figure S2. Accuracy of genomic prediction throughout the growth season. Heritability was calculated on an entry-mean basis. RMSE increased throughout development due to the increased scale of plant height throughout development.

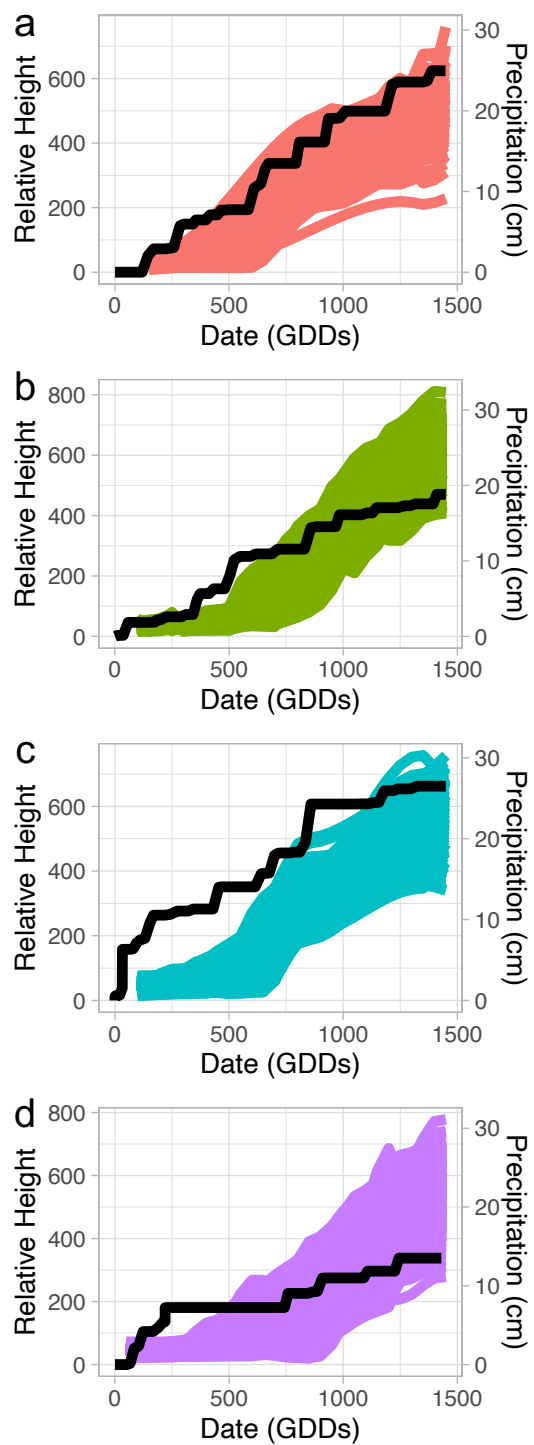


Figure S3. Plant height growth curves and precipitation accumulation throughout the growth season. The LOESS growth curves for all plots within each year on the left y-axis and the accumulation of precipitation throughout the growing season in black on the right y-axis. (a) 2018 (b) 2019 (c) 2020 (d) 2021.

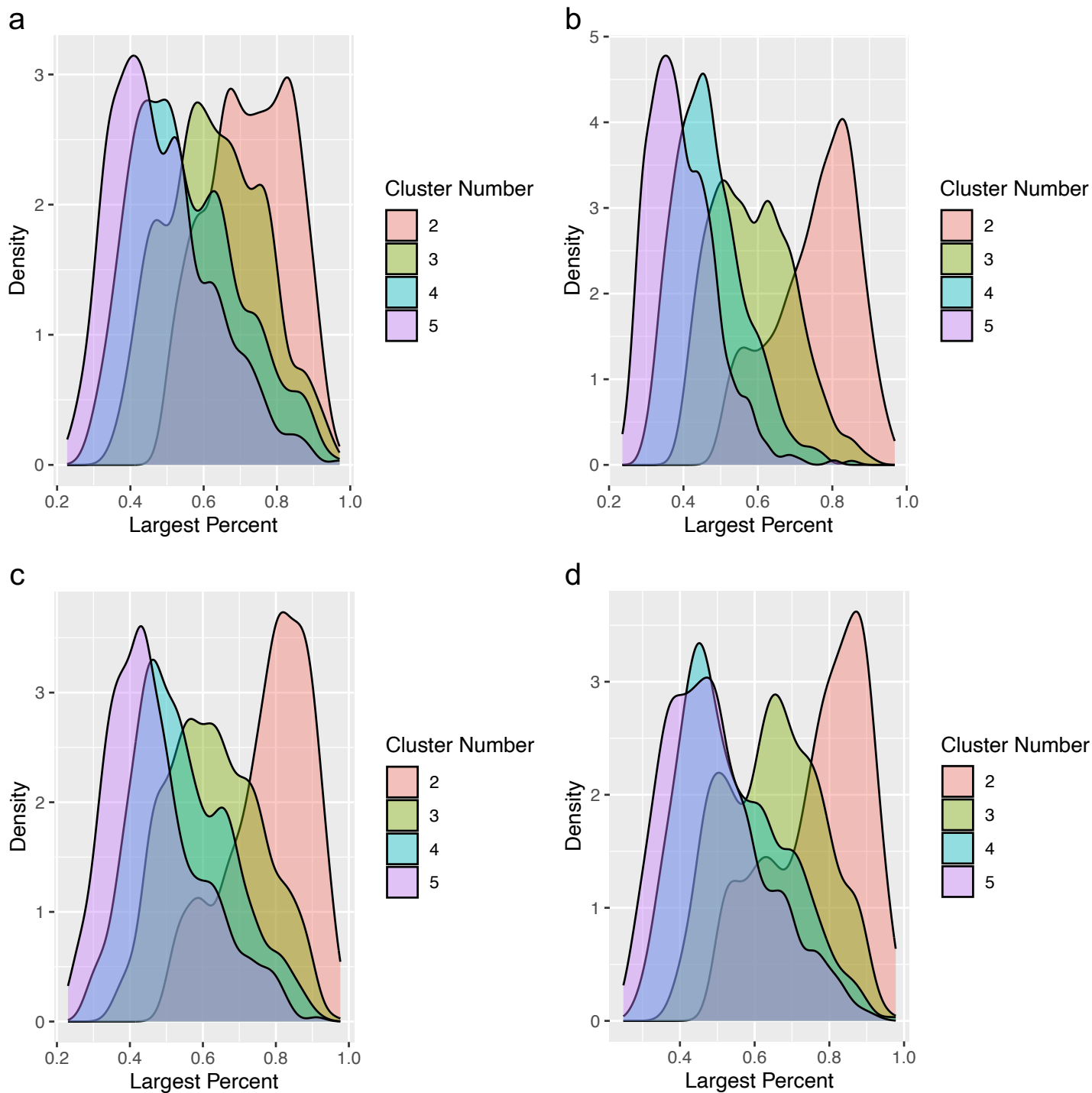
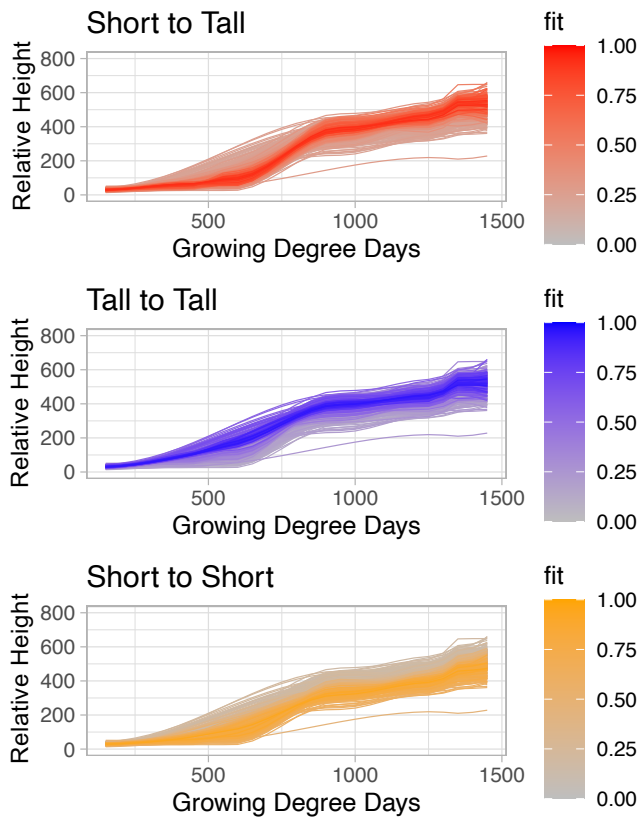
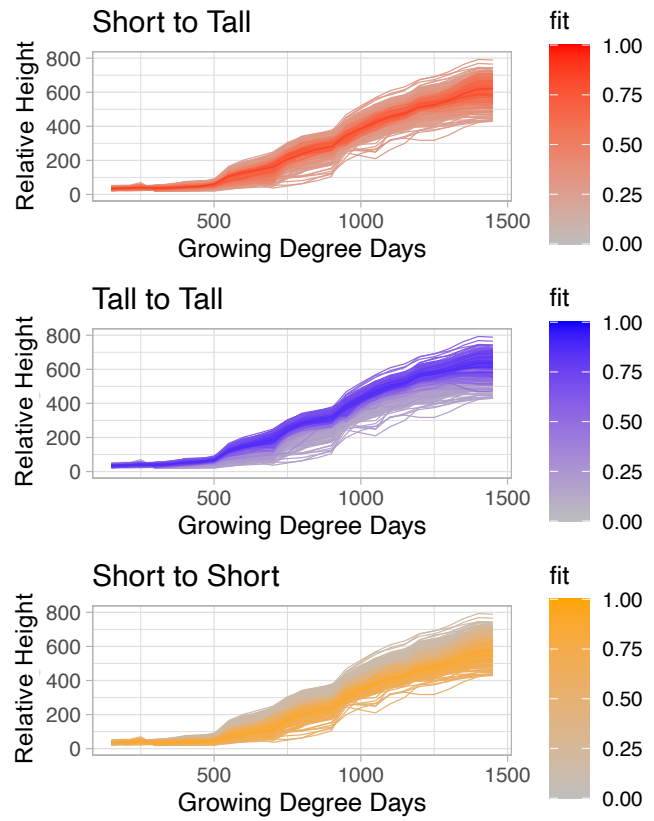


Figure S4. Density of largest percentages for determination of number of clusters to use for Fuzzy c-means clustering. The density of the largest value of wellness of fit for each curve to any single cluster when the curves are broken into 2, 3, 4, or 5 clusters with the value of the largest percentage of fit on the x-axis. These graphs were used to determine how many clusters fit the data best. (a) 2018 (b) 2019 (c) 2020 (d) 2021.

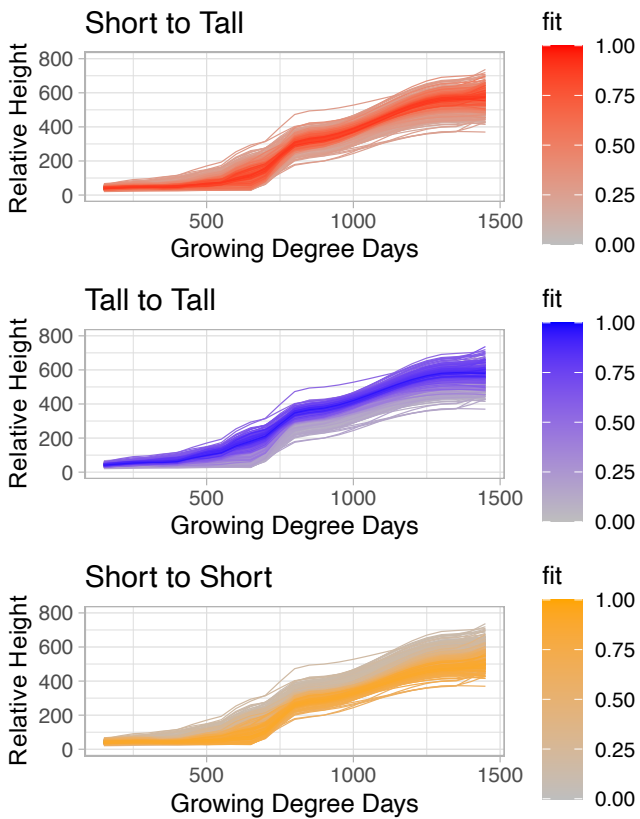
2018



2019



2020



2021

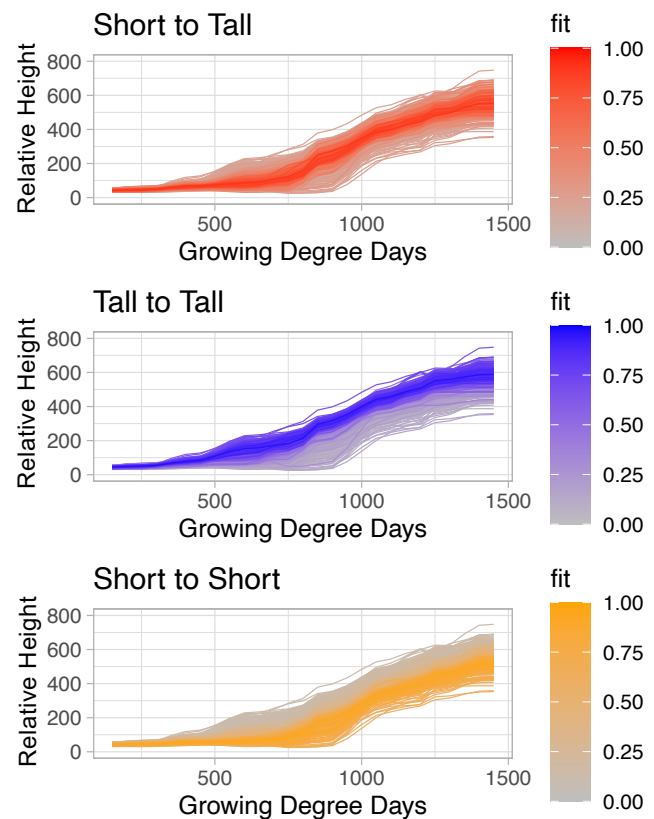


Figure S5. Fuzzy c-means clustering of plant height growth curve values. Fuzzy c-means clusters of LOESS growth curves with shading equating wellness of fit for each curve into the specified cluster separately for each year.

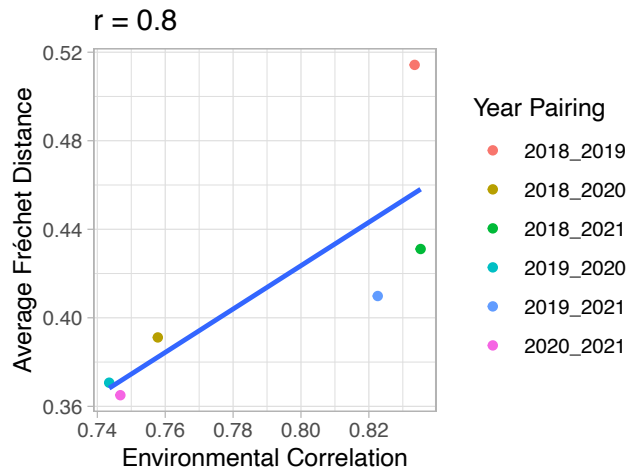


Figure S6. Pearson correlation between environmental similarity and average Fréchet distance. Fréchet distance between years for all genotypes with data in both years were averaged within each pair of years. Environmental correlations were calculated based on daily values for 18 weather parameters from planting to approximate flowering at 90 days after planting.

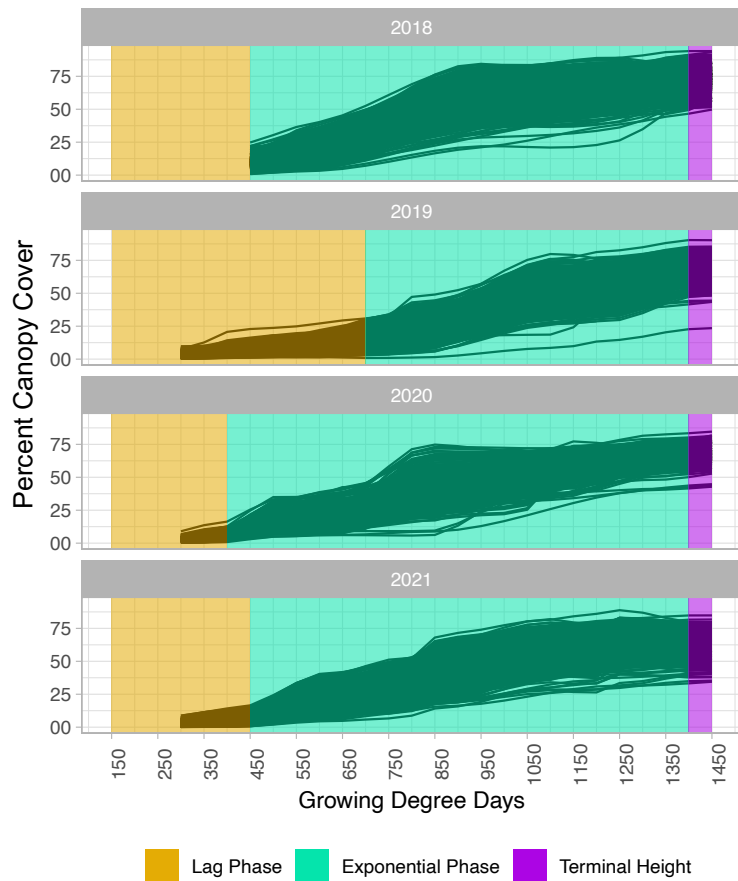
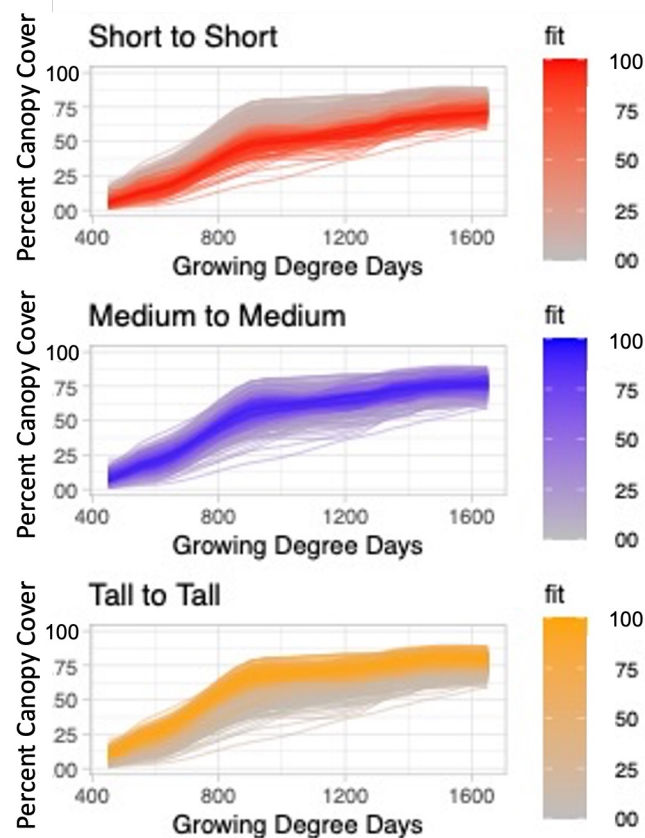
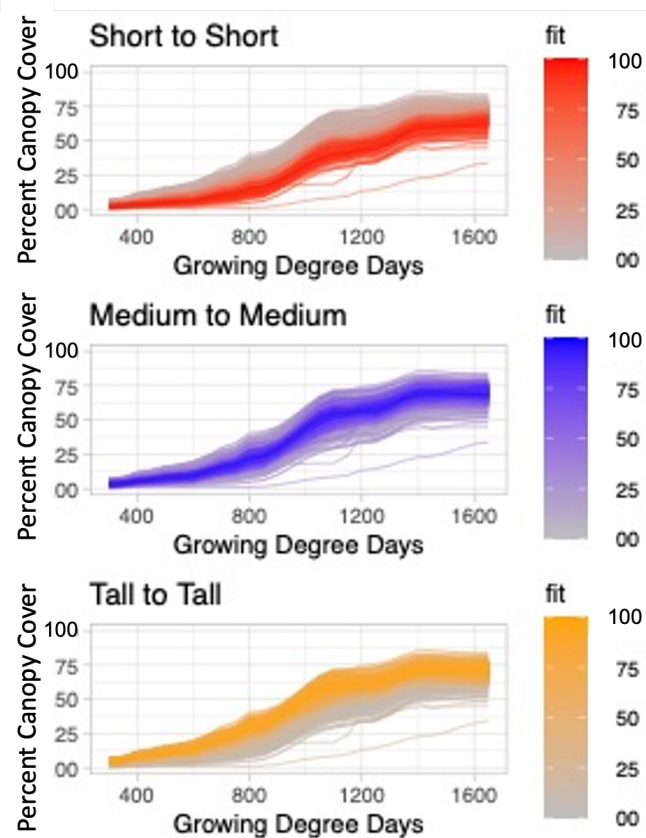


Figure S7. Temporal canopy cover growth curves. LOESS curves of percent canopy cover broken into three phases of the growing season based on the performance of genotypes in each year.

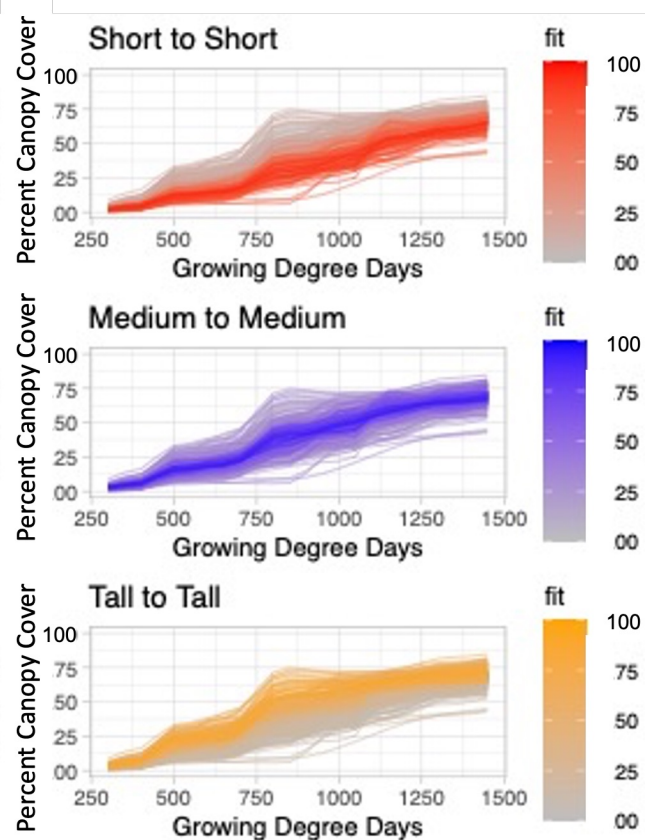
2018



2019



2020



2021

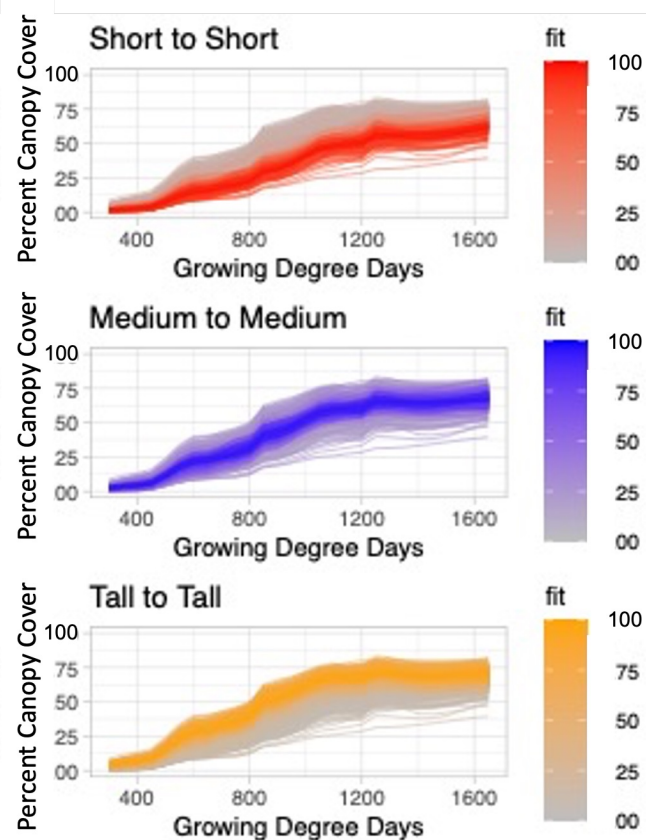


Figure S8. Fuzzy c-means clustering of canopy cover growth curve values. Fuzzy c-means clusters of LOESS growth curves with shading equating wellness of fit for each curve into the specified cluster separately for each year.

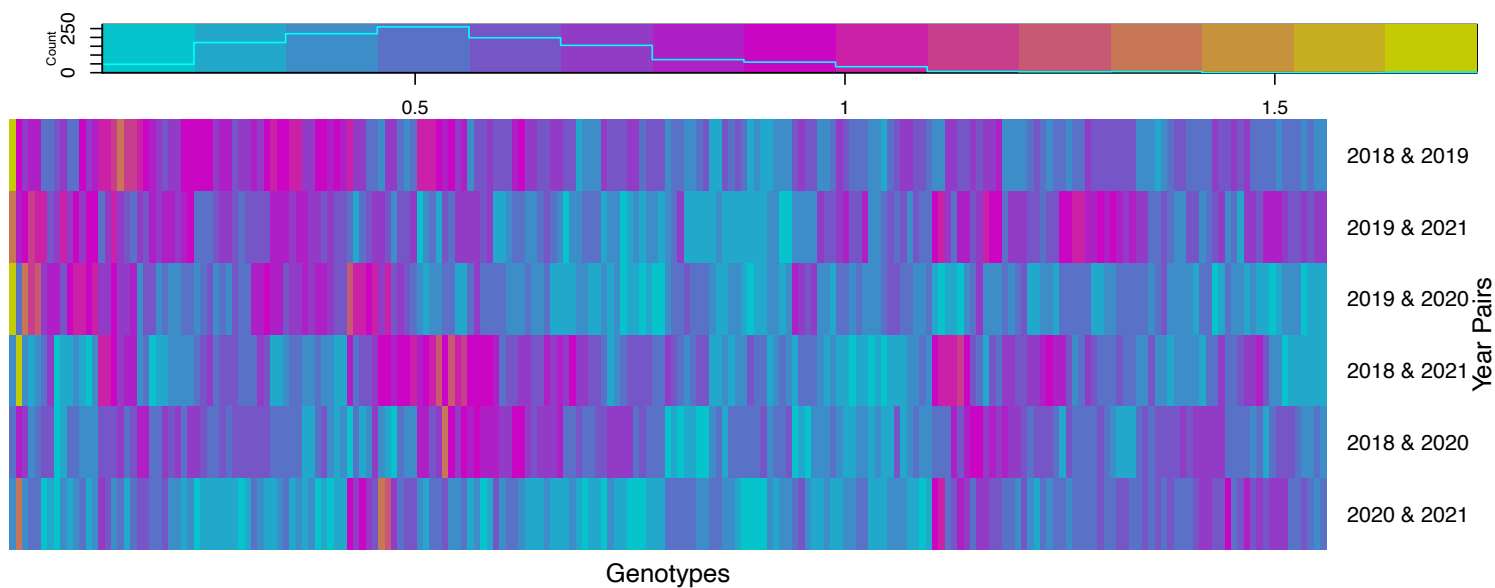


Figure S9. Fréchet distances for each genotype present in all four years. Pairwise Fréchet distance values comparing canopy cover growth curves of the same genotype across different years.

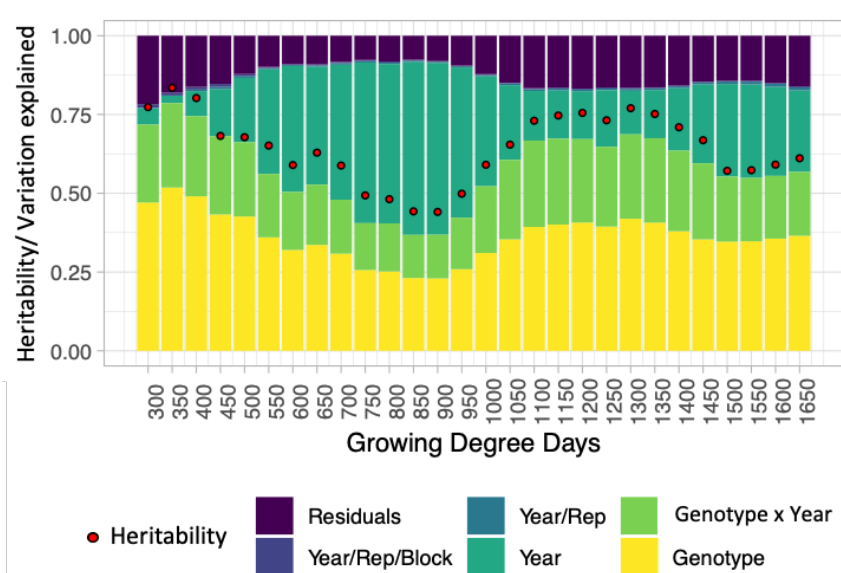
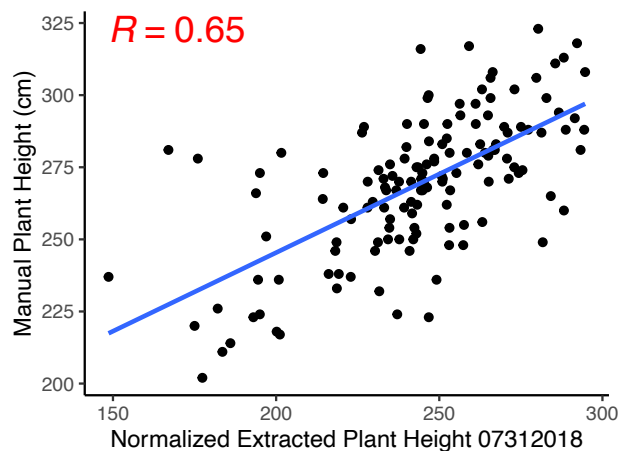
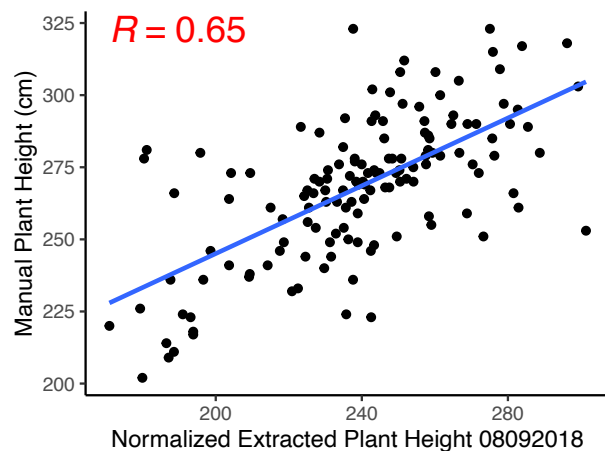
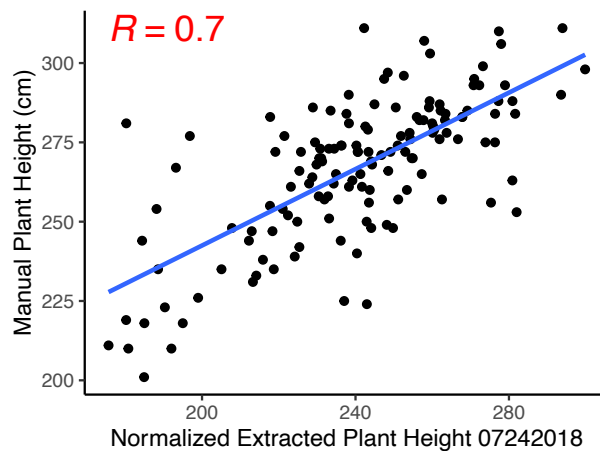
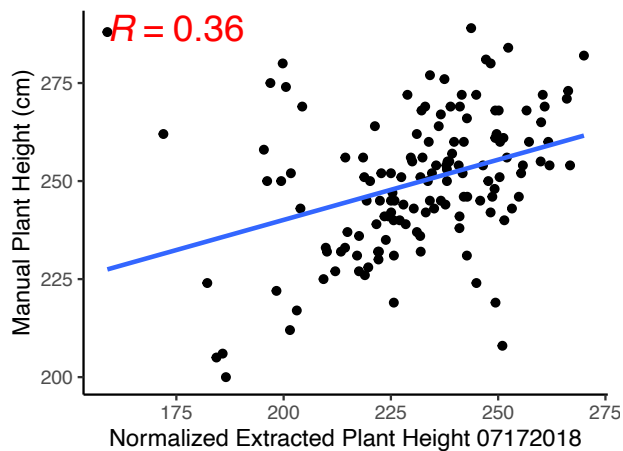
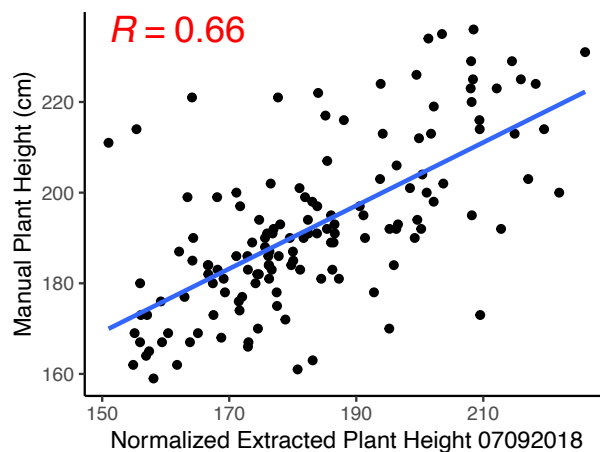
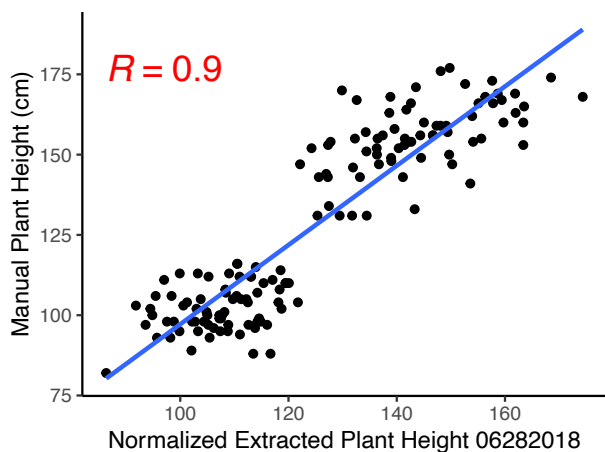
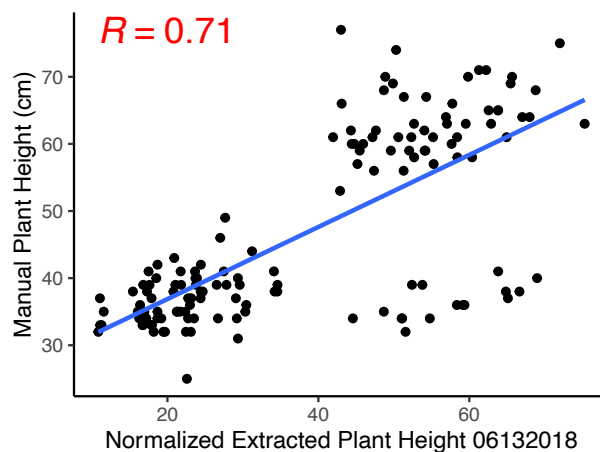
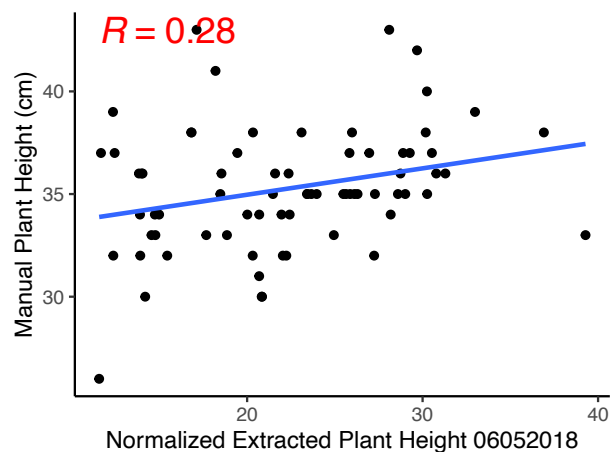
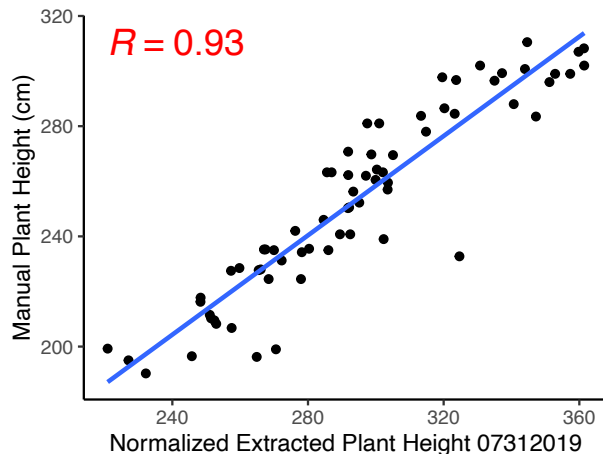
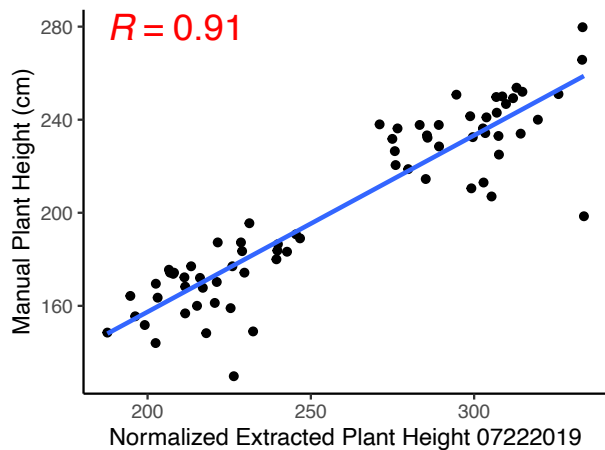
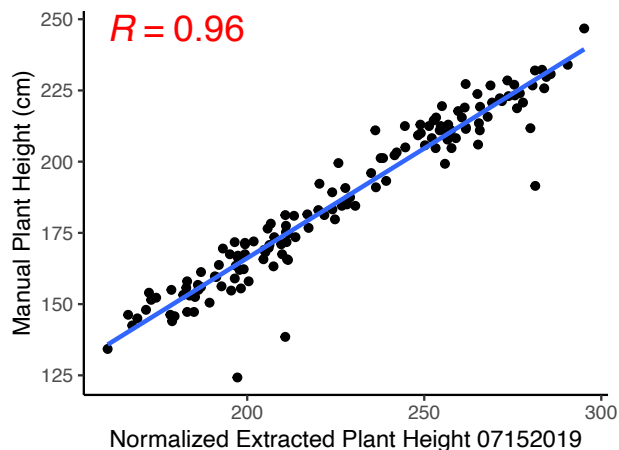
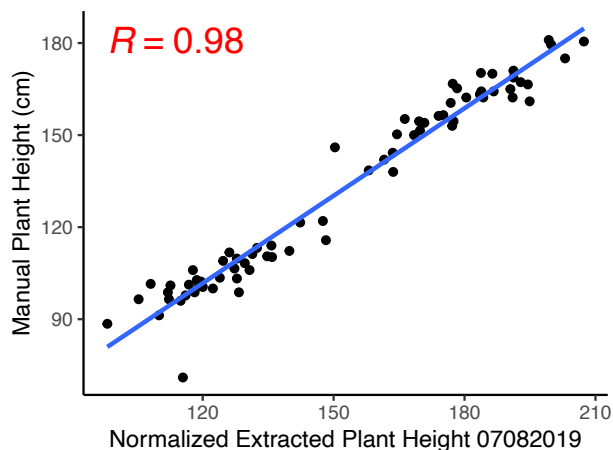
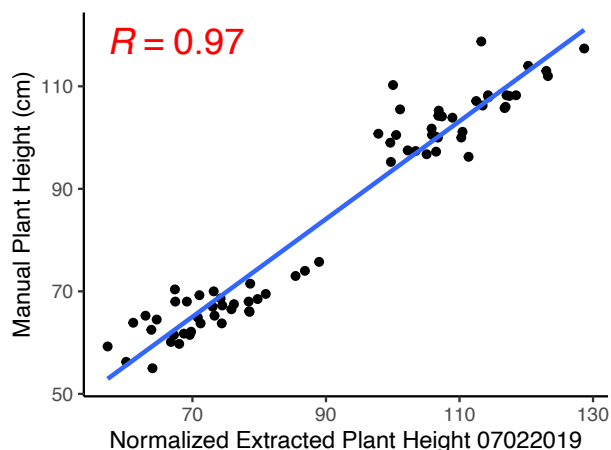
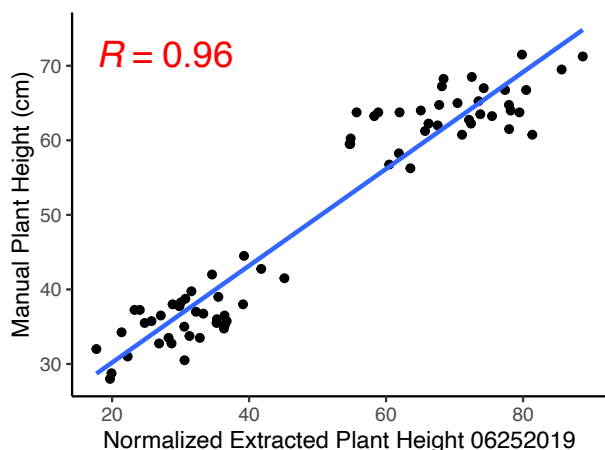
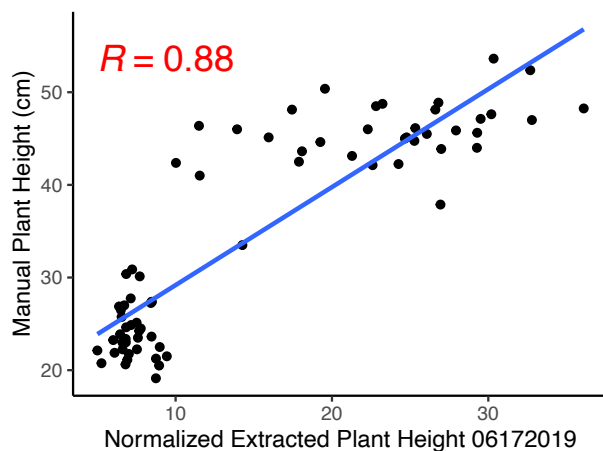
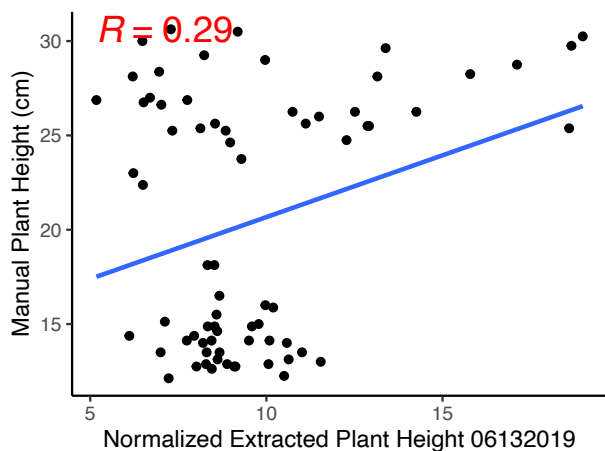
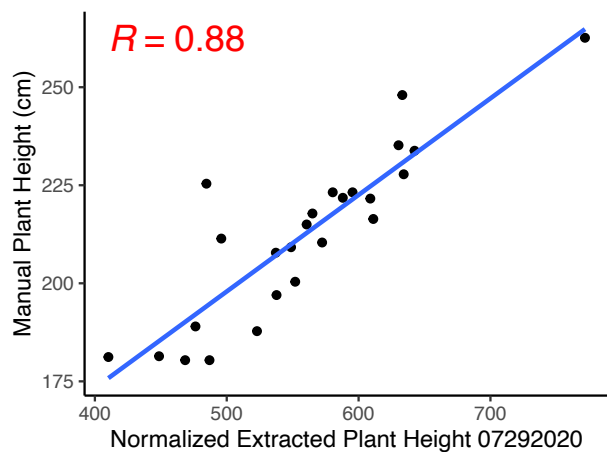
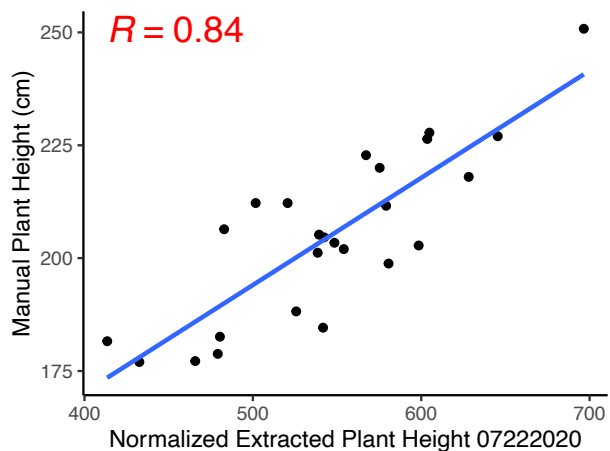
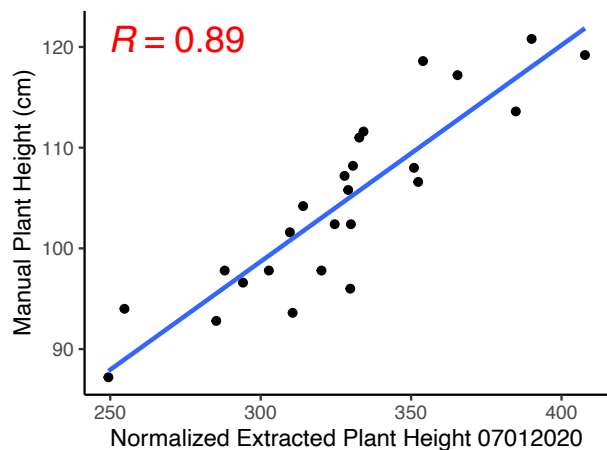
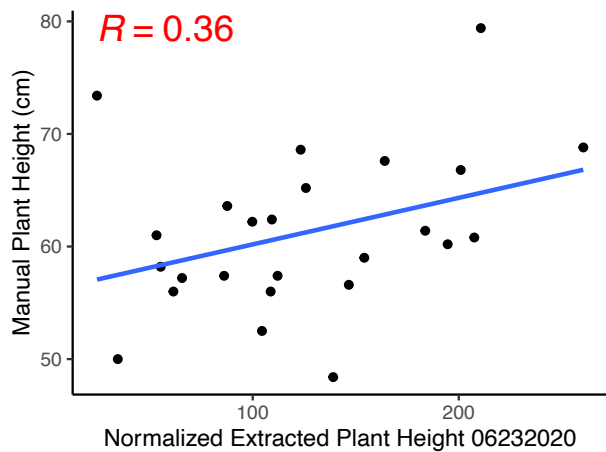
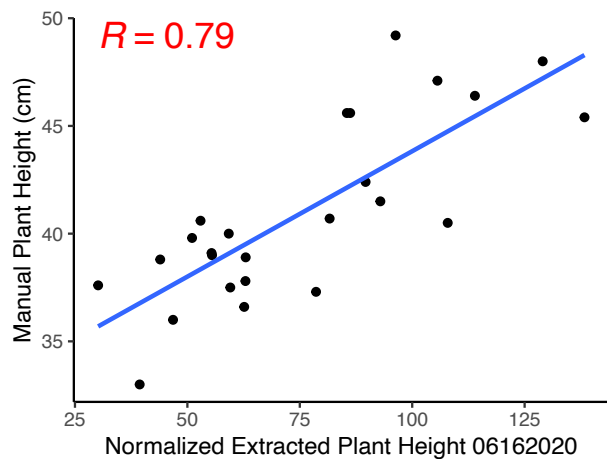
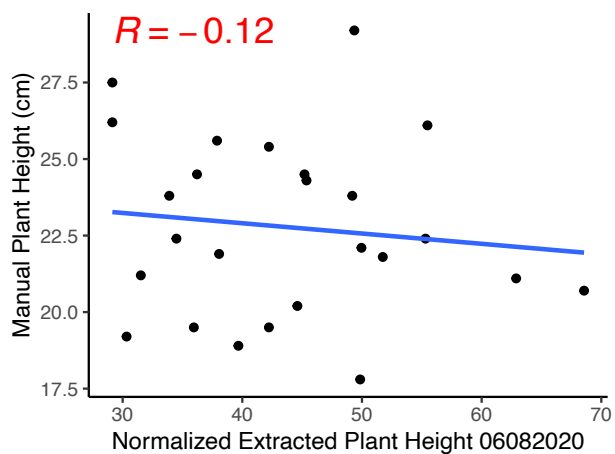
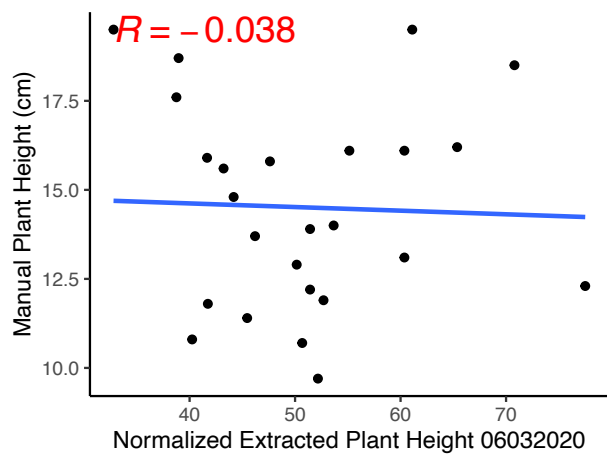
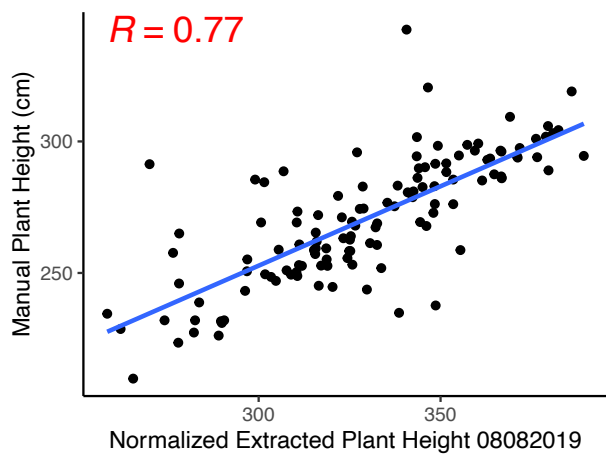


Figure S10. Temporal canopy cover analysis of variance. Percent variance explained and heritability at each canopy cover time point throughout the growing season.







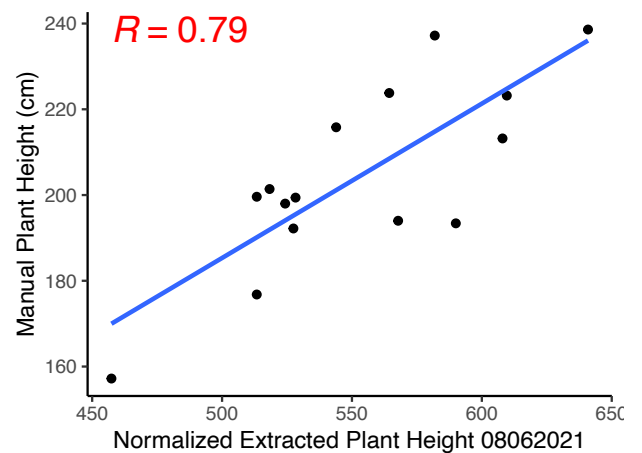
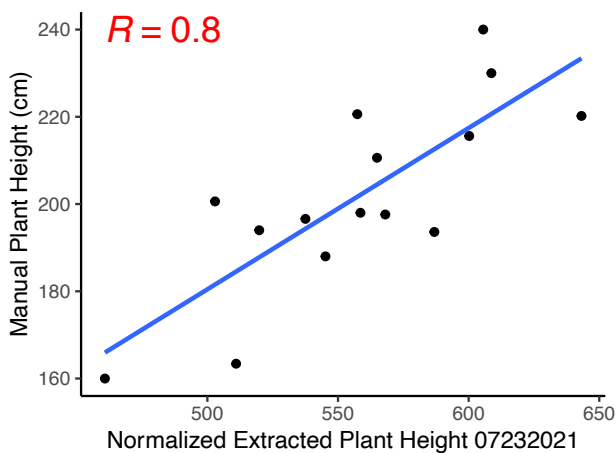
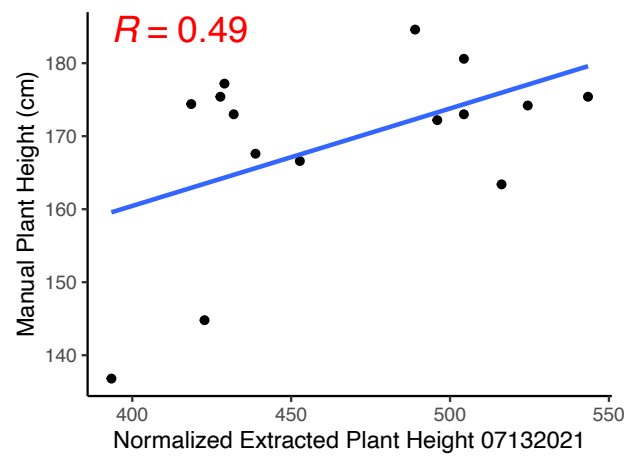
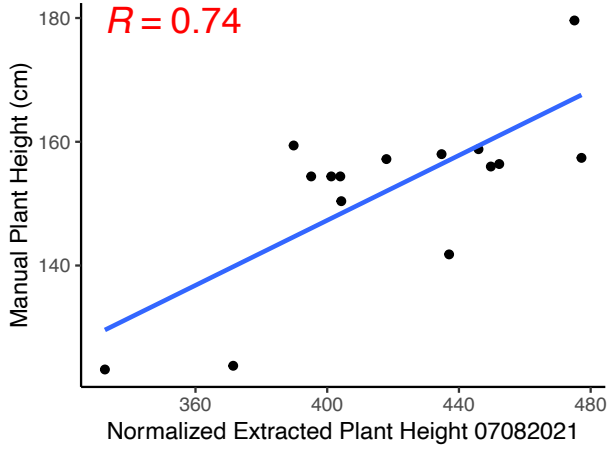
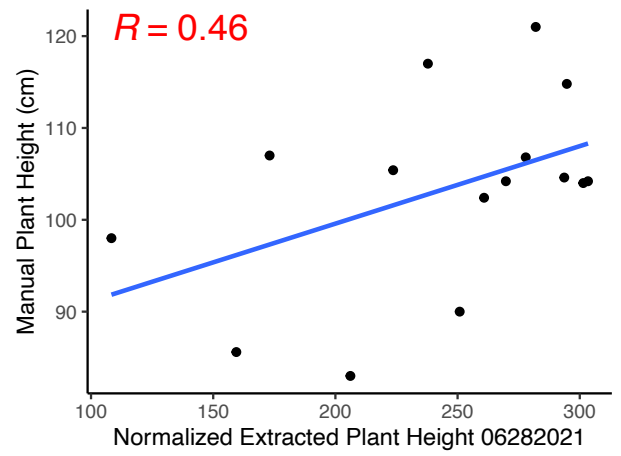
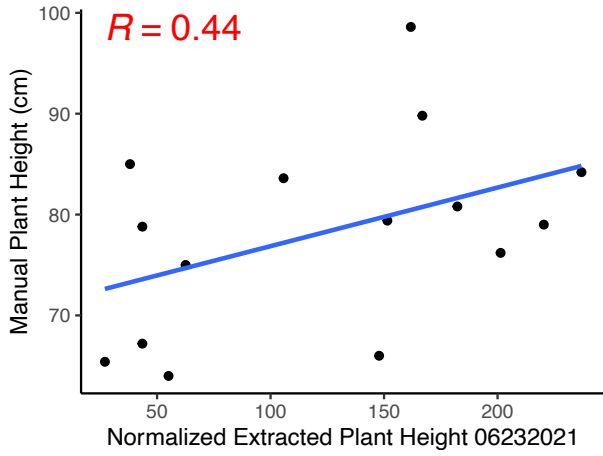
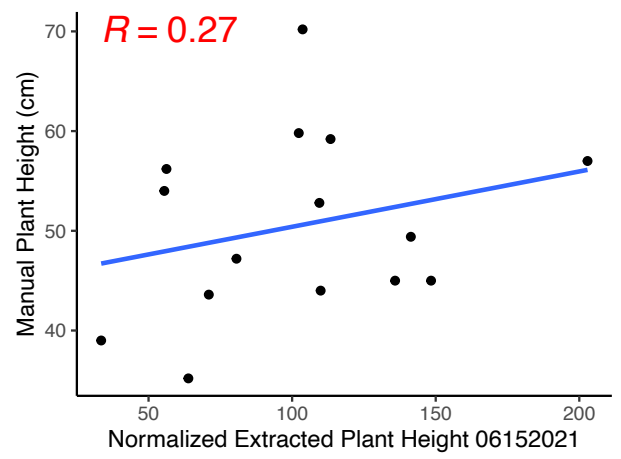
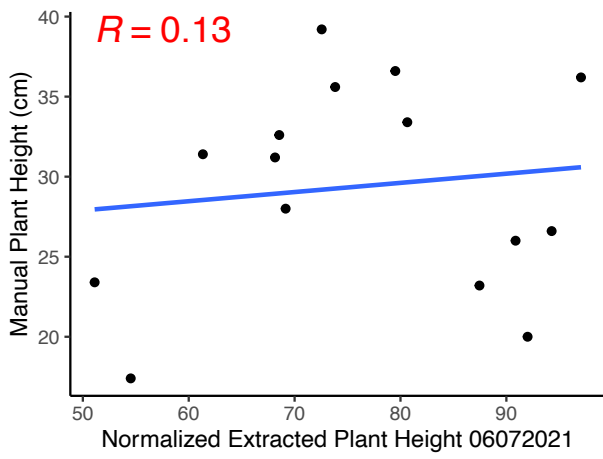


Figure S11. Validation of extracted plant height. Pearson correlation of normalized mean extracted plot plant height to mean manual plot plant height measurements. The date of each flight is indicated in the x-axis label in the form MMDDYYYY.

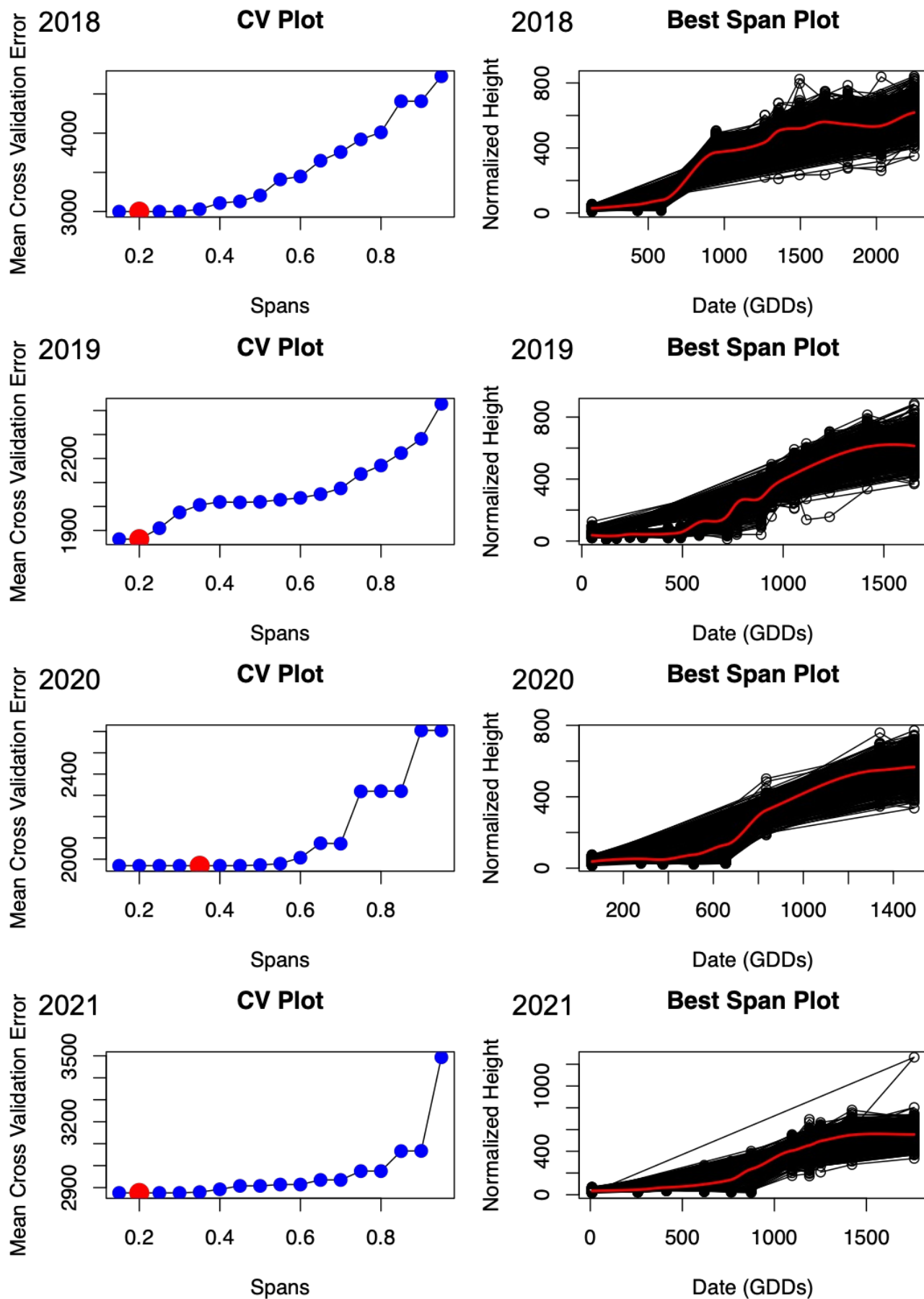


Figure S12. Span choices for LOESS curve fitting each year. Mean cross validation error for each possible span from 0.15 to 0.95 for each year with the best span for each year red (left plots). Growth rates with loess curves fit using the best span (right plots).